



Field monitoring and validation of vortex-induced vibrations of a long-span suspension bridge



Hui Li ^{a,*}, Shujin Laima ^a, Qiangqiang Zhang ^a, Na Li ^b, Zhiqiang Liu ^b

^a School of Civil Engineering, Harbin Institute of Technology, Harbin 150090, China

^b CCCC Highway Consultants Co., Ltd, Beijing 100088, China

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ABSTRACT

To investigate full-scale wind-induced vibrations of a long-span suspension bridge with a central span of 1650 m, a long-term wind and wind effect monitoring system was created. The basic wind field characteristics along the span-wise direction of the investigated bridge were analyzed. It was found that the wind field along the span-wise direction was inhomogeneous. The full-scale wind pressure distribution around the lower surface of the twin-box girder was also obtained. From the power density functions (PSDs) of the fluctuating pressures, the vortex shedding frequency of the full-scale twin-box girder was determined. A field visualization test was performed, and the flow pattern around the lower surface was obtained. Thirty-seven vortex-induced vibration (VIV) events were observed during the monitoring period. The corresponding wind conditions and vibrations were analyzed in detail. In addition to the wind direction and inflow turbulence, it was found that the inhomogeneity of the wind field along the span-wise direction of the bridge is also a critical factor that affects VIVs of full-scale bridge. The VIVs from a section model test and the full-scale bridge were compared, and it was found that the vertical VIV amplitude of the section model was much smaller than that from the field monitoring results. Moreover, torsional VIVs appeared in the section model test, whereas it was not observed in the full-scale bridge.

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1. Introduction

Although vortex-induced vibrations (VIVs) of a bluff body have been widely investigated, they are still not completely understood due to the complexity of the nonlinear interaction between the fluid and motion of the bluff body. Because this problem extensively exists in actual structures and negatively affects their safety. Comprehensive fundamental studies have been performed (Bearman, 1984; Gabbai and Benaroya, 2005; Sarpkaya, 2004; Williamson and Govardhan, 2004). A rough definition of VIVs can be expressed as “when flow passes through a bluff body, vortex shedding occurs in the wake. If the vortex shedding frequency is close to the natural frequency of the bluff body, large-amplitude vibrations will occur”. However, it is well known that this simple definition cannot adequately describe VIVs. VIVs exhibit extremely complex fluid dynamic characteristics. In recent decades, most studies have primarily focused on a bluff body with a simple configuration, such as in the aforementioned studies, for example, circular cylinders and rectangular prisms. However, little is understood on the VIV mechanism and flow patterns for more

complicated configurations, for example, a stream-like box girder, which is extensively used in modern long-span cable-supported bridges. For long-span cable-supported bridges, three special characteristics significantly affect the VIVs of a bridge deck. First, as the span size increases, the modes of the bridge become more and more dense, and the corresponding damping becomes smaller. For example, the lowest vertical frequency of a suspension bridge, whose main span size is 1650 m, is 0.09766 Hz, and the frequency interval between modes is approximately 0.04 Hz. The vertical modal damping is less than 0.60% (Li et al., 2011). Second, as the span size increases, the wind field along the bridge may become inhomogeneous. Finally and most importantly, the flow field around the stream-like box girder is more complicated than that around a circular cylinder due to separation and reattachment phenomena. Moreover, to improve the aerodynamic stability for a super-long span cable-supported bridge, twin- or multi-box girders have been extensively used in actual bridges, such as the Xihoumen suspension bridge (main span: 1650 m, China), Hong Kong Stonecutters cable-stayed bridge (main span: 1018 m, China), Gwangyang suspension bridge (main span: 1545 m, Korea), and the Strait of Messina Bridge (main span: 3000 m, Italy). Due to the gap between the separated box girders, the mechanism behind VIVs becomes more complicated. The VIVs of twin- or multi-box girders have been investigated through wind tunnel tests (Diana

* Corresponding author.

E-mail address: lihui@hit.edu.cn (H. Li).

et al., 2006; Ge et al., 2011; Ge, 2011; Larsen et al., 2008; Xu, 2004; Zhang et al., 2008). Although the wind tunnel test is a powerful tool to investigate VIVs of long-span suspension bridges, it cannot simulate actual wind dynamics and structural dynamic characteristics due to the size effects in the wind tunnel, i.e., the characteristics of the natural wind field at the bridge site, high Reynolds numbers and structural damping. It is known that these factors significantly affect the VIVs of long-span cable-supported bridges. However, field monitoring techniques can overcome the limitations of wind tunnel tests and provide insights into VIV behaviors of full-scale bridges. Moreover, it also can validate the wind tunnel test results. Larsen et al. (2000) discussed the VIVs of the Great Belt East Bridge observed during the final phases of deck erection and surfacing of the suspended spans, and the guide vanes were designed and implemented to mitigate the VIV oscillations of Great Belt East Bridge. Frandsen (2001) measured the wind pressures and accelerations of the Great Belt East Bridge simultaneously. In the field measured period, some VIV events on the Great Belt East Bridge were observed. It was found that the cross-wind VIVs of the Great Belt East Bridge occurred in smooth flow and low wind speeds (approximately 8 m/s) with a direction nearly perpendicular to the bridge axis. The time-average pressure coefficients on the top trailing edge were near zero and the correlation of span-wise pressures became large when VIV was occurring. Fujino and Yoshida (2002) observed that the first vertical VIV mode of a ten-span continuous single steel box-girder bridge (the Trans-Tokyo Bay Crossing Bridge) occurred in the wind direction within $\pm 20^\circ$ to the transverse axis of the bridge and the wind velocity approximately 16–17 m/s. To control

the vertical mode vibrations of the bridge, a new type of tuned mass damper (TMD) was implemented in the girder, and effective vibration control was achieved. Although successful examples on the studies of VIVs of full-scale long-span bridges through short-term field monitoring have been presented as mentioned above, it is necessary to do comprehensive studies on VIVs of full-scale long-span bridges through long-term field monitoring. Fortunately, more and more wind and wind effect long-term monitoring systems have been implemented on full-scale long-span bridges, and many VIV events from actual bridges have been recorded. Thus, it is possible to create comprehensive studies on VIVs of full-scale long-span bridges. Based on the above viewpoint, the primary objective of this paper is to investigate wind field characteristics and the VIV behaviors of a long-span suspension bridge with separated twin-box girder based on long-term field monitoring results. The VIVs of this long-span suspension bridge had been investigated in a previous article (Li et al., 2011) based on 2-month (9/29/2009–11/30/2009) field monitoring results. However, in the following 4-year monitoring process, the wind and wind effect monitoring system has been expanded and upgraded, and the accumulated field monitoring results are much more abundant than the previous article.

The organization of the paper is as follows: in Section 2, the investigated bridge and corresponding wind and wind effect monitoring system are introduced. In Section 3, the wind characteristics at the bridge site are analyzed in detail. In Section 4, the flow field characteristics around the twin-box girder are presented. In Section 5, the VIVs of the full-scale twin-box girder are investigated, and finally, the conclusion is presented.



Fig. 1. Location of the investigated long-span suspension bridge.

2. Descriptions of the investigated bridge and wind monitoring system

The investigated bridge is an asymmetric suspension bridge with a 1650 m central main span and a 578 m side span, which is located in the eastern sea in China. The bridge crosses a narrow water way that lies between two islands, as shown in Fig. 1. The bridge deck is composed of two separated steel box girders with 6 m space width. The detailed geometry configuration of the deck section is shown in Fig. 2.

Due to located in a special geographical environment, the bridge not only suffers from extremely smooth winds at low velocities, but also strong Typhoons and seasonal winds at high velocities. It is well known that long-span bridges are sensitive to these winds, resulting in dramatic oscillations under these wind conditions. To monitor wind field characteristics, wind-induced vibrations of the bridge and determine their mechanism, a wind and wind effect monitoring system was implemented in 2009 and then extended and upgraded in 2010. Six Young Model 8100 three-dimensional ultrasonic anemometers (see Fig. 3(a)) were installed on the eastern and western lighting columns at the 1/4, 1/2 and 3/4 central spans of the bridge (see Fig. 4(a)) at a height of 6 m above the deck surface. (Note that because the locations of the anemoscopes are close to the main cable in the middle of the center span, the measured winds are disturbed by the main cables. Thus, the winds at the 1/2 span were not analyzed in this study.)

For the orientation of the anemometer, zero degree represents the wind blowing from the north tower to the south tower, i.e., the bridge axis and the wind degree increases in the clock wise direction. The parameters of Young Model 8100 anemometer with range of [0–40 m/s] are as follow: resolution of wind speed: 0.01 m/s; resolution of wind direction: 0.1° ; accuracy of wind speed: $\pm 1\%$; accuracy of wind direction: $\pm 2^\circ$. Twelve force-balance uniaxial accelerometers (see Fig. 3(b), measurement range of ± 2.0 g, sensitivity of 2.7 V/g, and frequency band of (DC ~120 Hz), were installed at the 1/4, 1/2, 3/4 central spans and 1/2 side span. Each section had three accelerometers, two were installed on west side to measure the lateral and vertical accelerations of the bridge deck, and the other was installed on east side to measure vertical accelerations. The locations of the accelerometers are shown in Fig. 4(b).

Additionally, wind pressure field characteristics around the bridge deck are critical in determining wind loads and understanding the mechanism of wind-induced vibrations. Therefore, a number of wind pressure sensors were installed on the surface of the bridge deck at various sections. At the beginning of monitoring, 72 diffused silicon wind pressure sensors (CY200FA1P) were installed at the bridge surface at six sections (S1, S2, S3, S4, S5, and S6). Detailed information can be found in Li et al. (2011). Due to the zero-drift problem, these pressure sensors were removed. Recently, another 60 silicon capacitive-type pressure sensors (P993, as shown in Fig. 3(c)) were installed on the lower surface

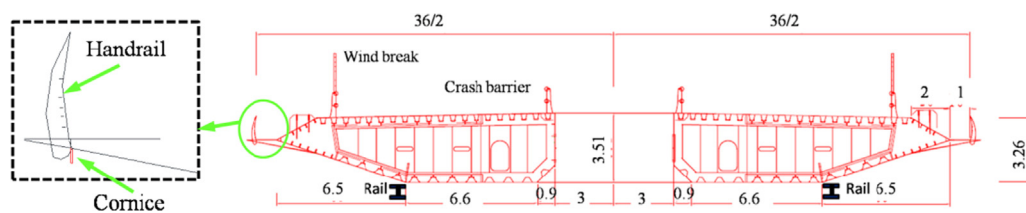


Fig. 2. Configuration of the twin-box girder.

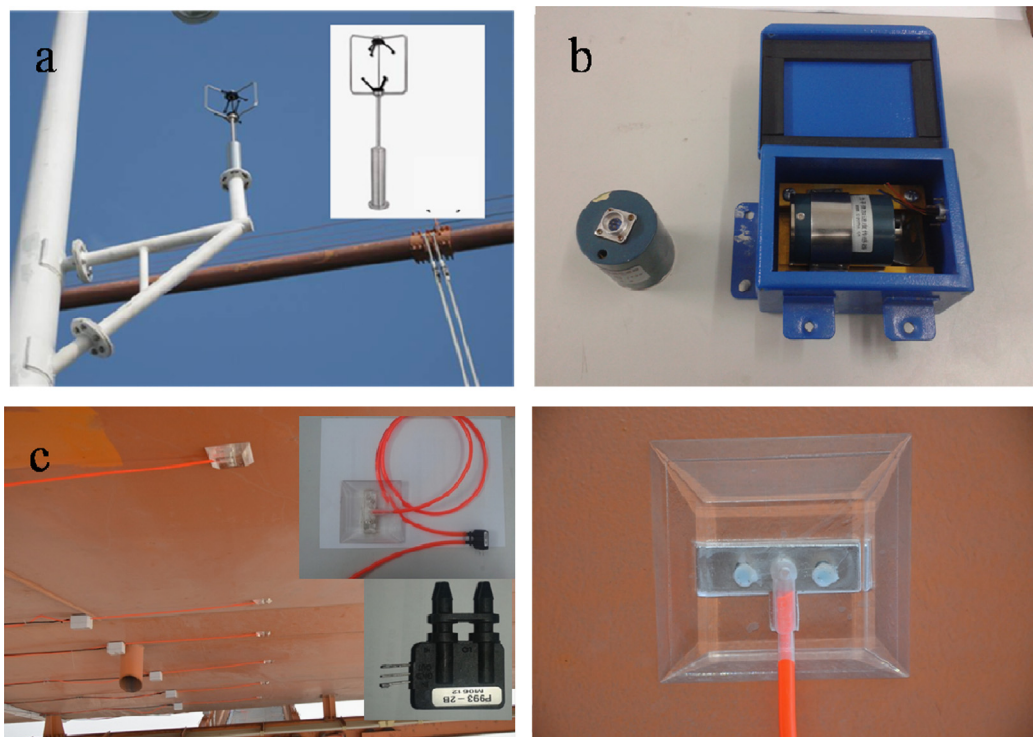


Fig. 3. Sensors used in the wind and wind effect monitoring system: (a) anemometer, (b) force-balance accelerometer, and (c) pressure measurement system.

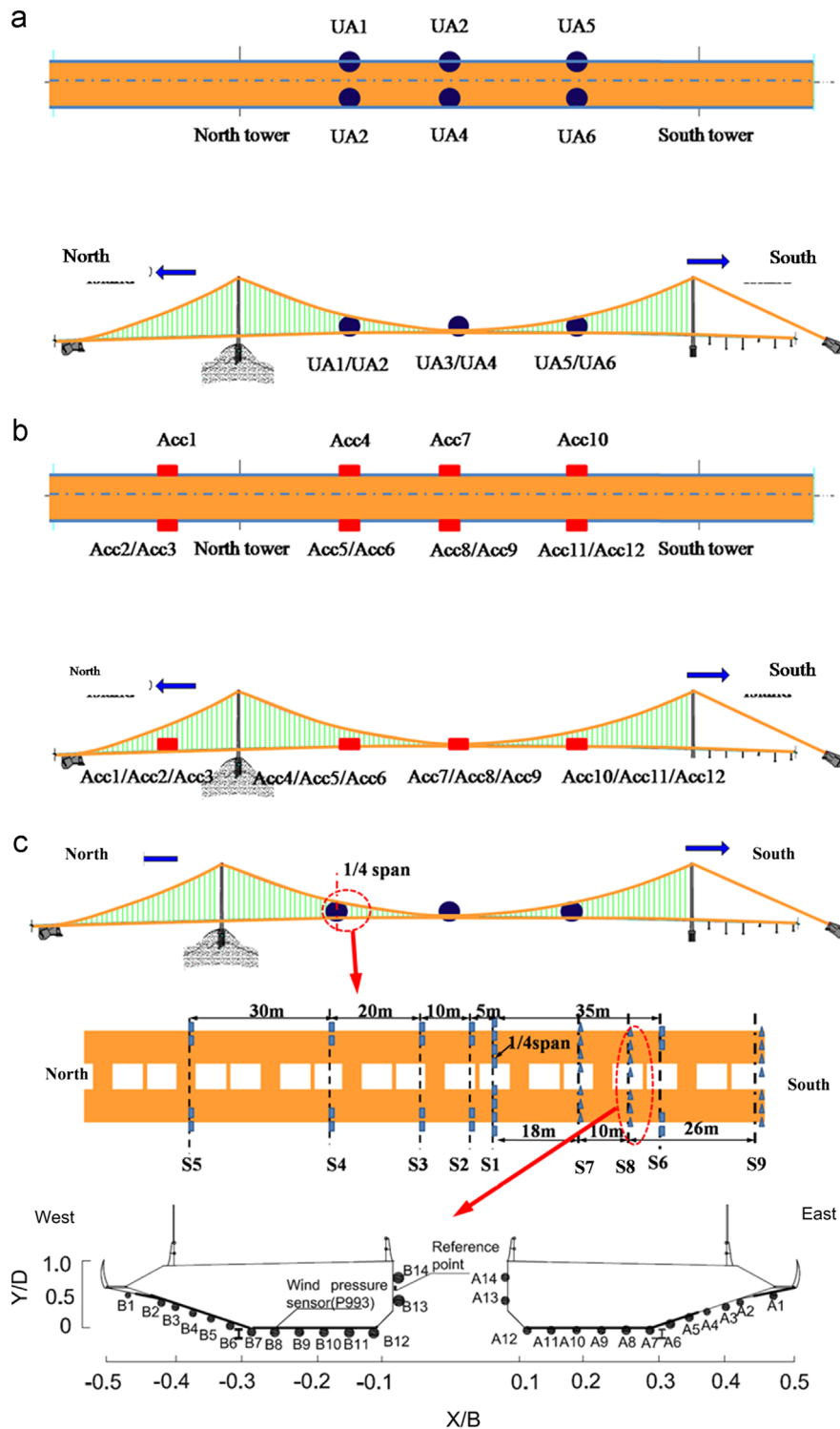


Fig. 4. Sensor locations used in the wind and wind effect monitoring system: (a) anemometer, (b) accelerometer, and (c) pressure sensor.

of the bridge deck at sections S7, S8 and S9. The locations are shown in Fig. 4(c). The P993 pressure sensor is a type of differential pressure PCB mount sensor with a small range (produced by Kavlico Company). The range of the P993 sensor used in this monitoring system is ± 498.0 Pa with a resolution of ± 0.5 Pa. The linearity of the P993 sensor is good, as shown in Fig. 5(a), and the upper bound of the frequency band is larger than 100 Hz (as shown in Fig. 5(b)). As Fig. 3(c) shows, the pressure measurement system includes four parts: the P993 sensor, organic glass plate with a pressure sensing

hole, 1-m sensing tube, and a reference tube. It is well known that the sensing tube will distort the pressure signal according to the size and length of the tube. Therefore, a sine calibration test was performed in laboratory. Fig. 5(c) shows the phase distortion vs. excited frequency curve. It can be seen from Fig. 5(c) that the phase lag is less than 5° when the excited frequency is less than 10 Hz. For the investigated bridge in this study, the concerned frequency range is between 0 and 1 Hz, and thus, pressure distortion can be ignored.

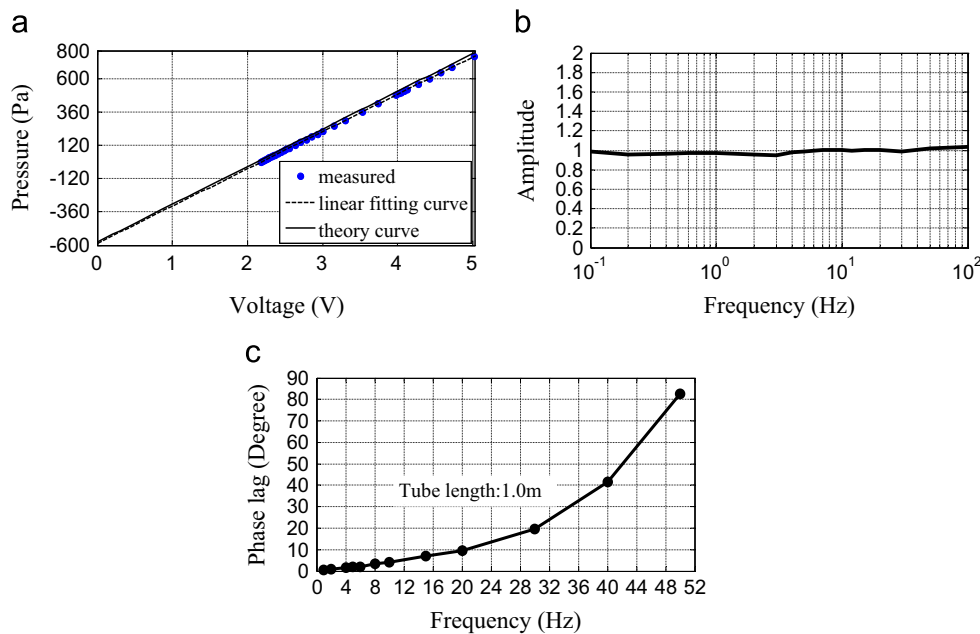


Fig. 5. P993 sensor characteristics: (a) relationship between sensing pressure and output voltage, (b) frequency characteristic, and (c) tube distortional effect.



Fig. 6. 1:25 section model test.

Field monitoring began on September 29th, 2009. In this paper, the data from April, July, September, and October in 2010 and April to September in 2011 were selected for the analysis. The sampling frequencies were set at 32 and 100 Hz for the ultrasonic anemometers and accelerometers, respectively. For the wind pressure sensors, the sampling frequency was set at 50 Hz.

To compare the full-scale aerodynamic characteristics and VIV response of the twin-box girder with the section model testing results, stationary and dynamic tests of a 1:25 section model were performed in the Joint Laboratory of Wind Tunnel and Wave Flume at the Harbin Institute of Technology, as shown in Fig. 6.

3. Wind characteristics at the bridge site during the summer season

Fig. 7, Tables 1 and 2 present the wind direction data, where 62.842% and 12.539% of the wind came from directions between $[60\ 120^\circ]$ and $[240\ 300^\circ]$ at the 1/4 span, respectively, and 57.574% and 12.854% of the wind came from directions between $[60\ 120^\circ]$

and $[240\ 300^\circ]$ at the 3/4 span, respectively. These results indicate that most of the wind was perpendicular or nearly perpendicular to the bridge during the summer season. It is known that perpendicular winds result in dramatic vibrations.

As shown in Fig. 1, the investigated bridge is located in a narrow asymmetric strait between islands. There are mountains on the islands, which induces the wind field close to the island is more easily disturbed. Due to the special geography, the wind field characteristics vary dramatically along the bridge span.

For wind directions between $[60\ 120^\circ]$ at the 1/4 span, the average 10-min mean wind speed, mean wind direction, and longitudinal turbulence intensity were 6.01 m/s, 89.68° , and 12.14%, respectively, whereas for the wind in the same direction range at the 3/4 span, these values are 5.340 m/s, 97.65° , and 22.69%, respectively. For wind directions between $[240\ 300^\circ]$ at the 1/4 span, the average 10-min mean wind speed, mean wind direction, and longitudinal turbulence intensity are 5.66 m/s, 266.97° , and 14.62%, respectively, whereas for the same wind direction range at the 3/4 span, these values are 5.36 m/s, 282.64° , and 21.03%, respectively. After further analysis on the turbulence distribution, it can be found that the range of turbulence intensity was extremely broad and varied dramatically along the bridge span. The proportion of wind at the 1/4 span with turbulence intensity less than 0.1 is 38.140%. However, at the 3/4 span, the proportion of wind with turbulence intensity less than 0.1 is 13.150%, as shown in Tables 3 and 4.

Comparing the wind characteristics at the 1/4 span and 3/4 span showed that clear discrepancies in wind characteristics along the bridge axis exist, particularly for the turbulence intensity of inflow. In general, the wind at the 1/4 span has a larger mean wind speed, lower turbulence intensity and is more perpendicular to the bridge than that at the 3/4 span. Fig. 8 shows the wind characteristics (10-min mean wind speed, 10-min mean wind direction, and turbulence intensity) at the 1/4 and 3/4 spans. The results shown in Fig. 8 and presented above indicate that the wind along the span-wise direction of the bridge was inhomogeneous. This result is different from the assumption of the current wind-induced vibration theory, which states that the wind field along the span-wise direction is homogeneous. Thus, the effects of inhomogeneity on wind-induced vibrations of a long-span bridge should be further studied in the future.

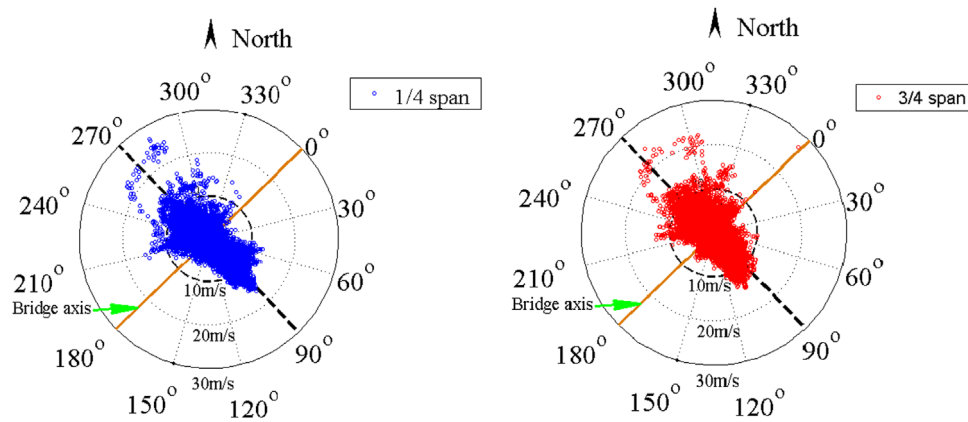


Fig. 7. 10-min mean wind speed and direction at the 1/4 and 3/4 spans.

Table 1

Joint PDF of the 10-min mean wind speed and wind direction at the 1/4 main span.

Wind direction	0–4 m/s	4–8 m/s	8–12 m/s	12–16 m/s	16–20 m/s	20–24 m/s	24–28 m/s	PDF (%)
0–30°	0.657	0.069	0.000	0.000	0.000	0.000	0.000	0.726
30–60°	4.281	0.378	0.204	0.003	0.000	0.000	0.000	4.866
60–90°	8.609	16.370	5.961	0.249	0.000	0.000	0.000	31.189
90–120°	7.616	14.897	8.411	0.726	0.003	0.000	0.000	31.654
120–150°	2.064	0.471	0.003	0.000	0.000	0.000	0.000	2.538
150–180°	3.153	0.366	0.000	0.000	0.000	0.000	0.000	3.519
180–210°	3.183	0.975	0.042	0.003	0.000	0.000	0.000	4.203
210–240°	2.079	2.805	0.486	0.066	0.000	0.000	0.000	5.436
240–270°	2.424	3.342	1.029	0.243	0.018	0.048	0.000	7.103
270–300°	1.881	2.214	0.837	0.255	0.006	0.039	0.045	5.277
300–330°	1.476	0.534	0.075	0.072	0.012	0.000	0.000	2.169
330–360°	0.912	0.345	0.048	0.018	0.000	0.000	0.000	1.323
PDF (%)	38.334	42.765	17.096	1.635	0.039	0.087	0.045	100

Table 2

Joint PDF of the 10-min mean wind speed and wind direction at the 3/4 main span.

Wind direction	0–4 m/s	4–8 m/s	8–12 m/s	12–16 m/s	16–20 m/s	20–24 m/s	24–28 m/s	PDF (%)
0–30°	1.023	0.123	0.003	0.000	0.000	0.000	0.000	1.149
30–60°	8.090	0.231	0.018	0.000	0.000	0.000	0.000	8.339
60–90°	5.325	9.776	0.759	0.012	0.000	0.000	0.000	15.872
90–120°	10.562	23.158	7.781	0.201	0.000	0.000	0.090	41.703
120–150°	2.823	0.564	0.009	0.000	0.000	0.000	0.000	3.396
150–180°	2.378	0.486	0.000	0.000	0.000	0.000	0.000	2.724
180–210°	2.874	0.336	0.006	0.000	0.000	0.000	0.000	3.216
210–240°	2.181	1.794	0.237	0.015	0.000	0.000	0.000	4.227
240–270°	2.226	2.982	1.062	0.180	0.021	0.033	0.000	6.504
270–300°	2.409	2.952	0.792	0.129	0.018	0.048	0.003	6.351
300–330°	1.896	0.897	0.282	0.129	0.015	0.030	0.000	3.249
330–360°	2.307	0.900	0.048	0.012	0.000	0.000	0.006	3.273
PDF (%)	43.953	44.199	10.997	0.678	0.054	0.111	0.009	100

Table 3

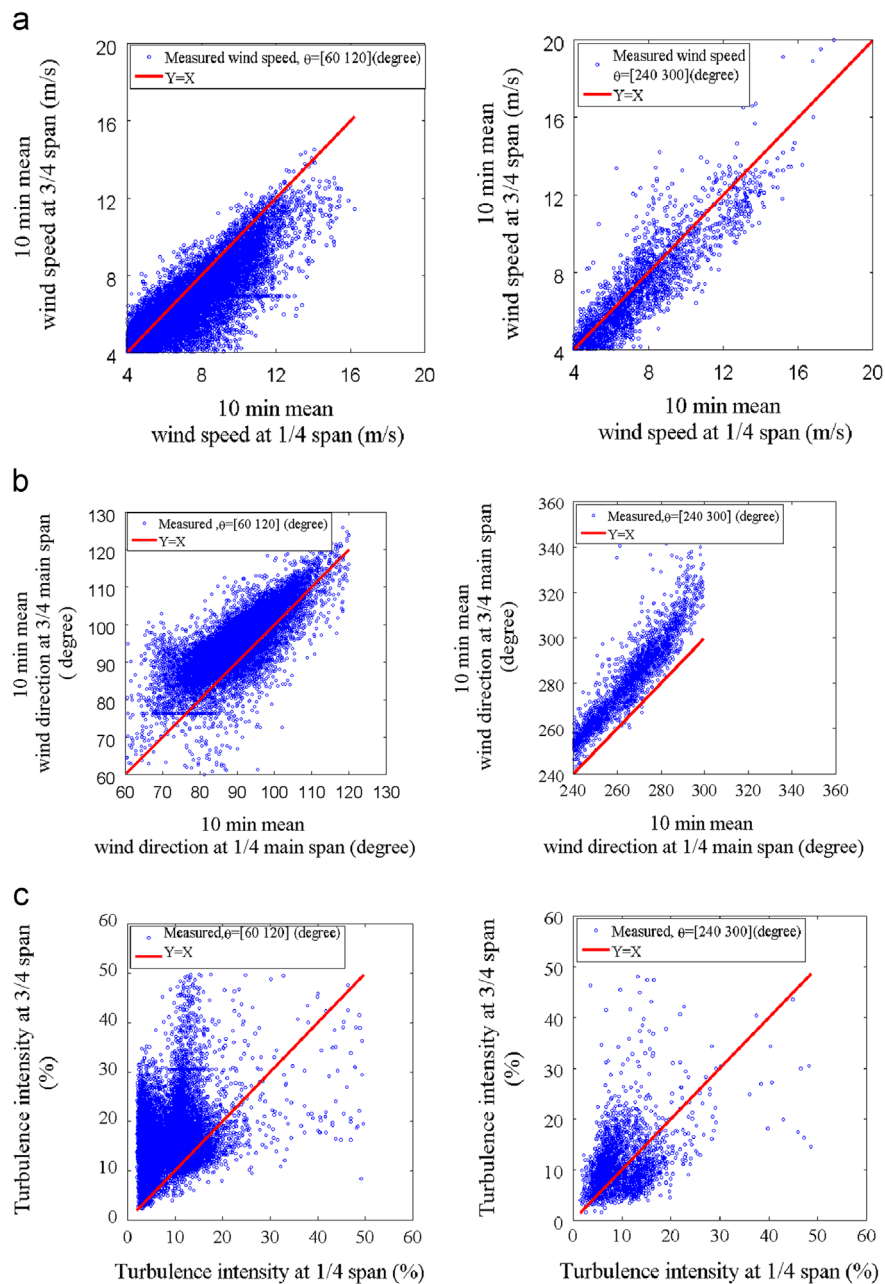
Joint PDF of the turbulence intensity and wind direction at the 1/4 main span.

Wind direction	0–2 (%)	2–4 (%)	4–6 (%)	6–8 (%)	8–10 (%)	10–12 (%)	12–14 (%)	> 14 (%)	PDF (%)
0–30°	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.532	0.532
30–60°	3.218	0.000	0.000	0.012	0.009	0.093	0.158	1.203	4.694
60–90°	0.000	0.000	0.000	0.166	3.274	10.483	7.992	9.867	31.801
90–120°	0.016	6.977	6.117	4.940	3.775	3.001	2.296	4.969	32.085
120–150°	0.000	0.006	0.096	0.356	0.415	0.306	0.201	1.037	2.417
150–180°	0.000	0.000	0.016	0.050	0.096	0.180	0.192	2.580	3.113
180–210°	0.000	0.000	0.000	0.000	0.000	0.000	0.000	4.196	4.196
210–240°	0.000	0.000	0.000	0.000	0.003	0.037	0.266	5.155	5.458
240–270°	0.022	0.306	0.907	0.944	0.0700	0.653	0.866	2.679	7.076
270–300°	0.000	0.136	0.801	1.309	0.869	0.514	0.040	1.262	5.294
300–330°	0.000	0.000	0.040	0.080	0.300	0.381	0.254	1.037	2.092
330–360°	0.000	0.003	0.046	0.059	0.105	0.074	0.068	0.888	1.244
PDF (%)	3.255	7.429	8.023	7.930	9.545	15.721	12.695	35.402	100

Table 4

Joint PDF of the turbulence intensity and wind direction at the 3/4 main span.

Wind direction	0–2 (%)	2–4 (%)	4–6 (%)	6–8 (%)	8–10 (%)	10–12 (%)	12–14 (%)	> 14 (%)	PDF (%)
0–30°	0.000	0.000	0.000	0.000	0.0032	0.016	0.001	0.776	0.804
30–60°	7.363	0.000	0.000	0.000	0.000	0.006	0.006	0.910	8.285
60–90°	0.000	0.000	0.000	0.000	0.201	1.829	2.853	11.140	16.024
90–120°	0.000	0.370	0.871	1.411	1.761	3.444	6.942	28.460	43.258
120–150°	0.000	0.026	0.201	0.383	0.329	0.329	0.287	1.628	3.182
150–180°	0.000	0.000	0.035	0.144	0.179	0.144	0.160	1.870	2.531
180–210°	0.000	0.000	0.000	0.000	0.000	0.000	0.000	3.128	3.128
210–240°	0.000	0.000	0.000	0.000	0.003	0.035	0.214	3.910	4.162
240–270°	0.016	0.073	0.479	0.613	0.712	0.795	0.788	2.729	6.204
270–300°	0.003	0.121	0.629	1.072	1.114	0.954	0.619	1.820	6.332
300–330°	0.000	0.006	0.010	0.032	0.099	0.243	0.393	2.215	2.997
330–360°	0.000	0.000	0.051	0.137	0.268	0.399	0.424	1.812	3.093
PDF (%)	7.382	0.597	2.276	3.792	4.669	8.193	12.696	60.396	100

**Fig. 8.** Relationship of wind characteristics between the 1/4 span and 3/4 span: (a) 10-min mean wind speed, (b) 10-min mean wind direction, and (c) turbulence intensity.

4. Flow field characteristics of the twin-box girder

The wind pressure characteristics, wind pressure distribution, vortex-shedding frequency, and rough vortex shedding pattern around the bridge deck can be obtained by analyzing the monitored wind pressure. The wind pressure coefficient is defined as follows:

$$C_p = \frac{p - p_0}{\frac{1}{2} \rho U_0^2} \quad (1)$$

where p_0 is the 10-min mean pressure at the reference position, which is located at the wall of the gap in this study, as shown in Fig. 4, U_0 is the 10-min mean velocity of inflow, p is the wind pressure at the measured position, ρ is the air mass density, and C_p is the wind pressure coefficient.

Fig. 9 shows the time histories of the monitored wind pressure.

Fig. 10(a) and (b) presents the pressure coefficient distributions of mean wind pressures and RMS wind pressures on the lower surface at section S8 of the bridge deck respectively. One hundred and seventy samples, which share similar wind conditions, were selected to be analyzed. The corresponding wind conditions are as follows: average 10-min mean wind speed of 170 samples was 11.68 m/s, average 10-min mean wind direction was 272° , and average turbulence intensity was 6.90%. In this figure, the solid line denotes the mean value of 170 samples, and the lines with the dots and circles are the lower and upper bounds, respectively. Note that VIVs did not occur in all of the analyzed samples. As shown in Fig. 10, the mean pressure coefficient reached its maximum suction value around the windward corner and then became extremely small rapidly. It is interesting that there is a small region on the windward side where the pressure shows an adverse pressure gradient, indicating that the flow separates in this the

region. As shown in Fig. 2, there exists a cornice, which can induce flow separation. Due to the influence of the cornice and handrail, the pressure in the windward area fluctuates dramatically. Furthermore, the pressures around the windward and leeward corners also show strong fluctuations due to the rails.

For comparison, a wind tunnel test was conducted on a 1:25 scale section model in the wind tunnel at the Harbin Institute of Technology. The pressure coefficient distribution on this section model is also shown in Fig. 10. It can be seen that the wind pressure distributions obtained from the wind tunnel test are similar with those obtained from field monitoring; however, the wind tunnel test results had smaller mean pressures and less fluctuations. The discrepancies between the two results can be primarily attributed to two factors. One factor is that several detailed configurations were not completely simulated in the section model due to too small a scale, e.g., the cornice. The second factor is that the Reynolds number in the wind tunnel test is much smaller than that in the field.

To further investigate flow separation at the lower surface, a field visualization test was performed on the full-scale bridge. The field visualization system shown in Fig. 11 is comprised of three parts, a laser light sheet device with a maximum power output of 3 W, smoke generator and a CCD camera. Fig. 12(a) shows the smoke visualization results under a mean wind speed of 10.03 m/s, wind direction of 267° , and turbulence intensity of 6.60%, and no any VIV occurred. The figure shows that the flow separates from the lower surface of the rail, and the separation region is wide due to the large incident angle of flow around the rail. Moreover, when the flow separates, large coherent vortex structures appear simultaneously. The large coherent vortex structures directly travel across the lower side of the gap, which indicates that there are no alternating shedding Karman-like vortices in the gap.

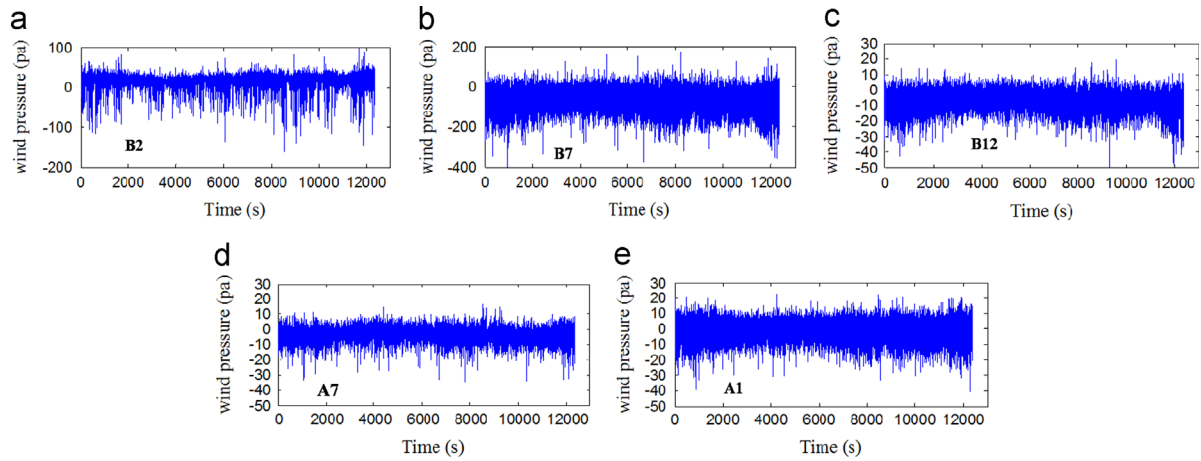


Fig. 9. Time history of pressure at different locations: (a–e) B2, B7, B12, A7, and A1.

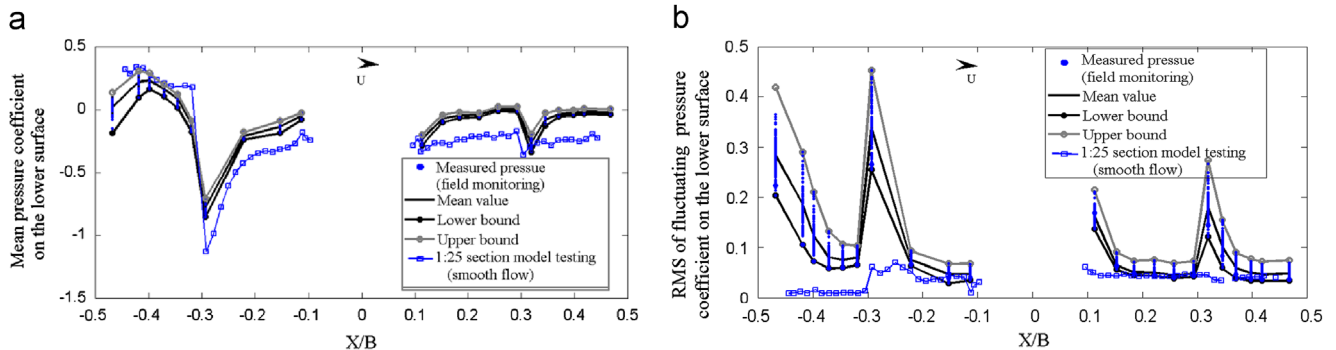


Fig. 10. Pressure distribution of the full-scale bridge and 1:25 section model: (a) mean pressure and (b) RMS of fluctuating pressure.

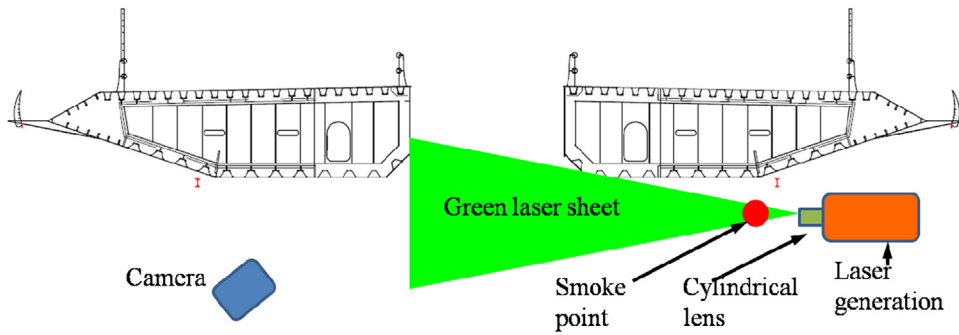


Fig. 11. Schematic of the flow visualization system used in the full-scale bridge model.

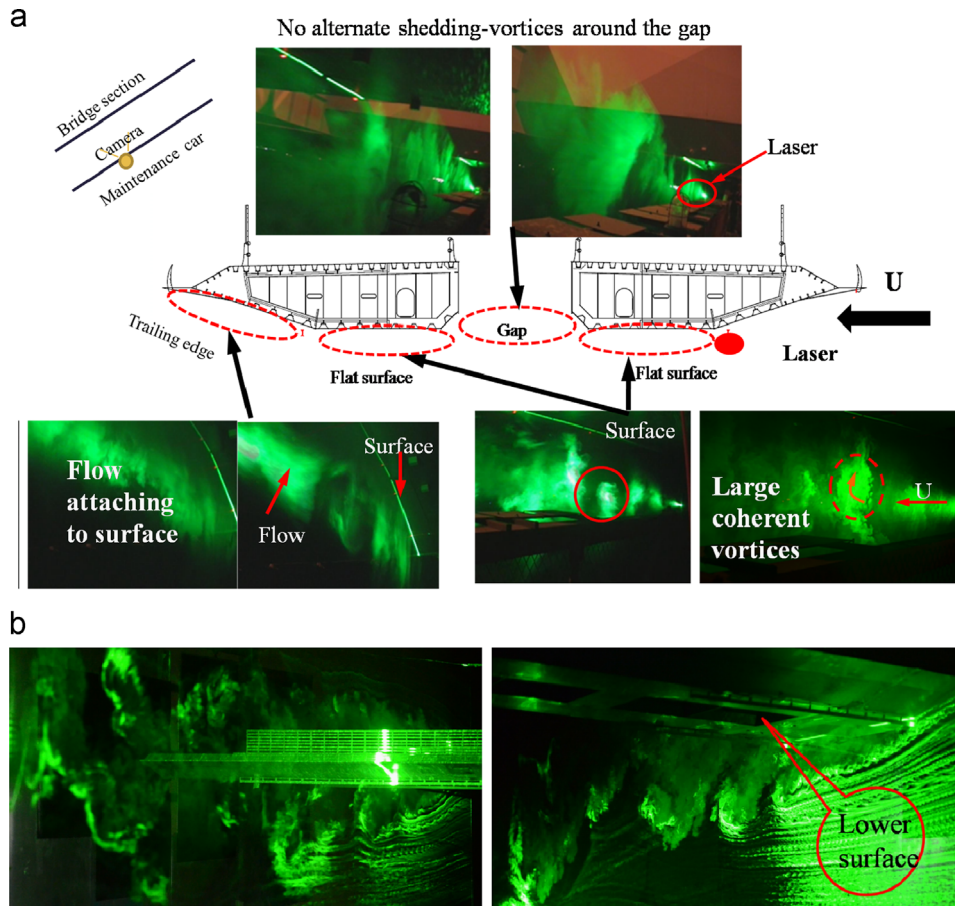


Fig. 12. Smoke visualization of the flow around the lower surface of the twin-box girder. (a) Field monitoring and (b) 1:25 section model test.

For comparison, flow visualization of the 1:25 section model is shown in Fig. 12(b). It can be seen that the flow around the trailing edge do not reattach on the surface of the section model, which indicated that the section model test overestimated flow separation around the lower surface of twin-box girder.

Fig. 13 shows the PSDs of pressures on the lower surface with the same wind conditions as the smoke visualization test, in which the VIV did not occur. It can be observed that most pressures on the lower surface did not have dominant frequency except at the rear of the downstream box girder. It is well known that, the period vortex shedding can induce the pressure on the surface of body having the identical period fluctuating character. Therefore the peak frequency of pressure is used to determine vortex shedding frequency. The relationship between inflow velocity and regular vortex shedding frequency in the tail area is shown in Fig. 14. It can be seen that the vortex shedding frequencies of

the full-scale twin-box girder obey Strouhal's law, which states that the vortex shedding frequency increases linearly as the inflow velocity increases. The average Strouhal number ($St = f_v D/U$, where St is the Strouhal number, f_v is the vortex shedding frequency in the rear of the downstream box girder, D is the center height of the twin-box girder, and U is the inflow velocity) of the full-scale twin-box girder is 0.1159. Note that the f_v used here is the vortex shedding frequency at the rear of the downstream box girder (see Fig. 13), not the dominant frequency determined from the integral of the pressure around the section surface, because it is hard to obtain this integral with field monitoring. Due to the larger separation of the flow than in full-scale bridge, as shown in Fig. 12, the Strouhal number of the 1:25 section model test ($St = 0.0892$, as shown in Fig. 14(b)) is much smaller than that of the full-scale bridge. According to the above pressure and flow visualization analysis, a schematic of the rough flow pattern

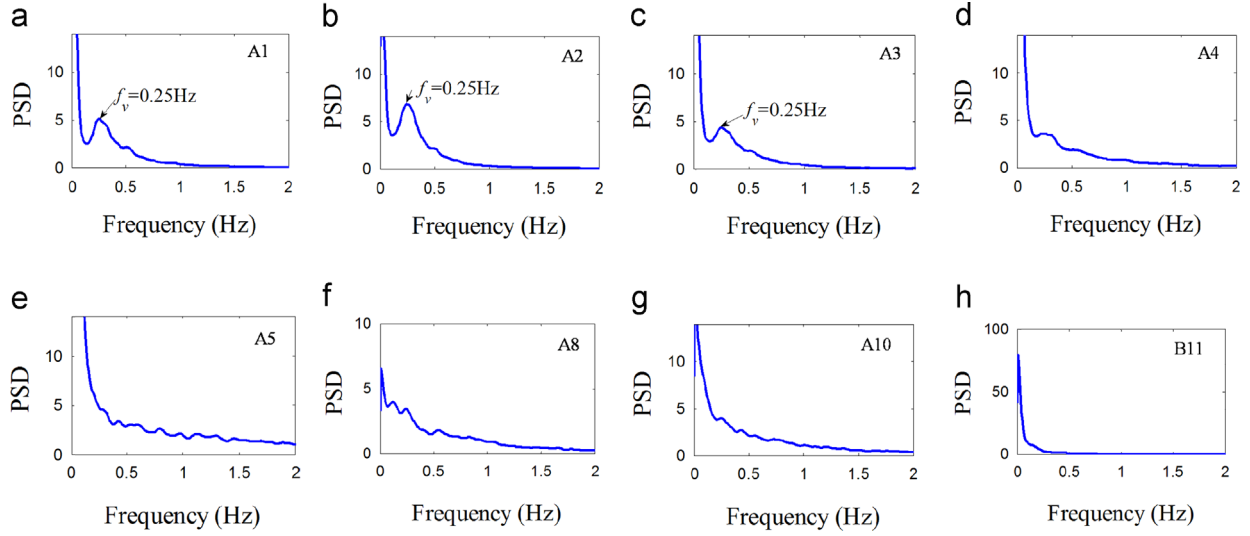


Fig. 13. Power density functions (PSDs) of pressures at typical locations under a mean wind speed of 10.03 m/s, wind direction of 267°, and turbulence intensity of 6.6% (no any VIV occurred in this wind condition): (a–h) A1, A2, A3, A4, A5, A8, A10, and B11 (as shown in Fig. 4, “A” is in the tail area for the wind direction of 267°).

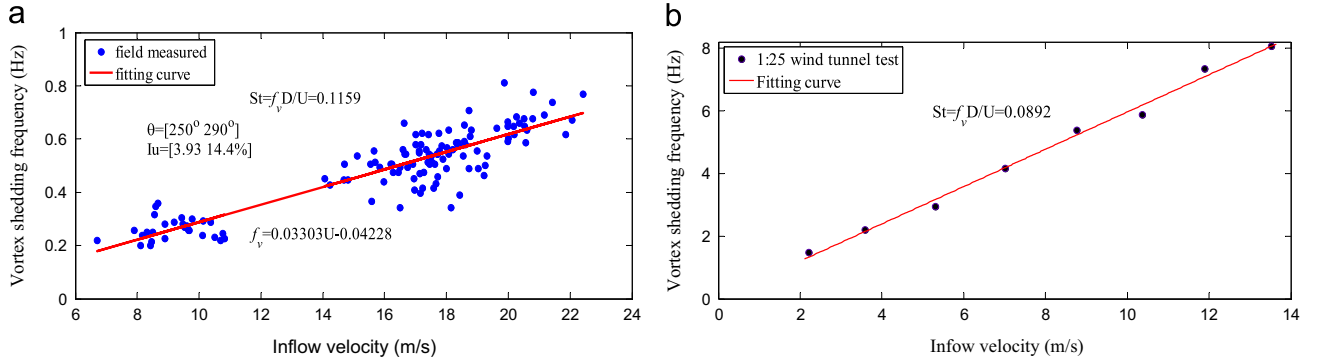


Fig. 14. Relationship between inflow velocity and vortex shedding frequency at the tail of the downstream box girder: (a) field monitoring and (b) 1:25 section model test.

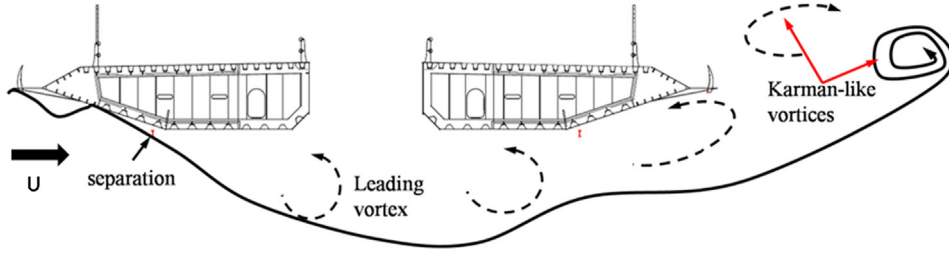


Fig. 15. Rough schematic of the flow pattern around the lower surface of the twin-box girders.

around the lower surface of the twin-box girder is shown in Fig. 15. The flow separates at the lower surface of the rail accompanied with large vortex shedding, and then the leading vortex propagates downstream and merges with vortices that are created at the trailing edge of the downstream box girder, finally forming the Karman-like vortices in the wake.

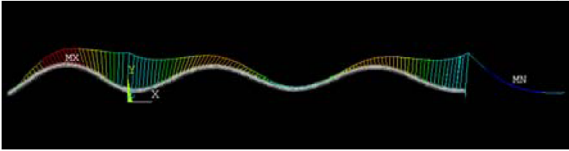
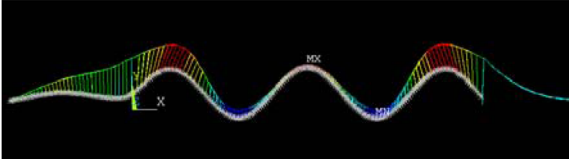
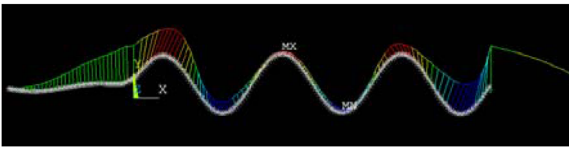
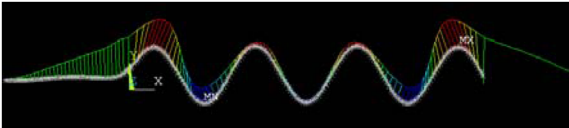
5. Vortex-induced vibrations of the twin-box girder

5.1. Wind conditions of the vortex-induced vibrations of full-scale bridge

Thirty-seven VIV events have been observed since 2009. Table 5 lists these VIV events in detail. The modal parameters

were obtained based on the stochastic subspace identification (SSI) method (Li et al., 2011) and FEA analysis. In Table 5 and in the following analysis, with the exception of the effects of inhomogeneity on the VIVs analysis, all wind parameters were obtained at the 1/4 span. Note that only vertical vortex-induced vibrations were observed. Four VIV modes with frequencies of 0.1831, 0.2295, 0.2734, and 0.3265 Hz were observed for this bridge. The maximum amplitude of displacement was 23.89 cm with a frequency of 0.1831 Hz, and the maximum amplitude of acceleration was 61.82 cm/s² with a frequency of 0.3265 Hz. Fig. 16 shows the acceleration time histories of two typical VIV events and corresponding PSDs. Fig. 17 presents the relationships of the RMS of displacement of the bridge deck and the corresponding wind conditions at the 1/4 span for all the VIV events that the displacement contained one single frequency only. From the

Table 5
VIV events of the bridge deck.

Sequence number	Modal parameters	Wind speed (m/s)	Wind direction (deg.)	Turbulence intensity I_u (%)	Max acc (cm/s^2)	Max dis (cm)	Duration (min)
1	Structural damping: 0.5%; natural frequency: 0.1831 Hz 	5.42–6.83	267–282	2.09–6.05	31.36	23.89	210
2		6.81–7.36	268–272	3.01–5.24	21.22	16.31	40
3		5.16–5.78	280–284	4.75–6.91	9.36	7.90	40
5		5.75–6.50	273–276	3.50–6.00	12.66	9.68	50
6		4.78–7.17	260–276	1.80–9.20	20.54	15.85	110
7		5.32–6.25	266–281	2.40–8.60	22.37	17.02	120
8		5.62–7.08	88–97	2.50–11.00	13.98	10.30	100
9	Structural damping: 0.51%; natural frequency: 0.2295 Hz 	8.00–8.50	275–265	4.00–5.01	30.10	14.43	150
10		7.52–7.64	264–276	2.22–4.10	10.34	4.37	30
11		7.33–7.91	94–99	2.83–3.35	9.64	3.87	40
12	Structural damping: 0.39%; natural frequency: 0.2762 Hz 	8.56–9.04	263–273	3.01–6.94	30.68	10.75	60
13		9.08–10.04	265–270	2.45–8.17	25.54	8.79	60
14		9.08–10.21	258–267	4.30–6.75	35.06	11.68	130
15	Structural damping: 0.42%; natural frequency: 0.3265 Hz 	11.20–12.71	272–277	4.20–6.10	44.06	11.89	40
16		9.73–10.68	260–265	3.23–5.88	30.78	9.61	40
17		10.18–11.56	94–98	2.19–3.92	18.47	6.48	150
18		9.75–10.44	93–94	4.90–7.07	13.46	4.64	50
19		11.59–11.90	93–94	3.08–5.65	13.86	7.08	40
20		10.75–11.16	92–96	2.34–4.52	17.11	5.54	100
21		9.96–10.67	105–106	2.54–4.53	13.85	3.64	30
22		10.43–11.42	89–93	3.29–7.44	12.30	4.02	70
23		10.44–11.33	252–262	4.60–5.69	50.89	14.78	50
24		10.20–12.00	260–275	3.50–6.01	61.82	17.10	170
25		10.92–12.64	260–265	3.70–5.70	40.18	11.39	60
26		10.53–11.20	97–102	2.50–4.60	8.46	2.42	130
27		11.05–11.48	94–103	2.10–2.90	12.47	2.87	30
28		9.78–11.51	88–99	3.60–9.90	25.10	5.05	60
29		10.30–10.86	93–99	2.30–4.70	8.40	2.21	90
30		11.46–11.95	91–98	2.60–8.60	6.95	1.89	30
31		10.00–11.39	91–95	4.30–11.10	8.74	2.29	30
32		11.65–12.57	96–102	2.50–3.60	8.91	2.26	60
33		11.24–12.17	94–98	2.30–4.70	13.01	3.03	40
34		9.87–11.49	94–98	2.20–3.70	8.15	2.12	90
35		9.82–10.71	90–95	2.90–8.10	6.09	1.67	40
36		10.48–11.08	90–95	4.30–8.10	15.49	3.70	30
37		9.74–11.35	273–271	6.72–8.94	24.96	5.64	150

figure, it can be concluded that most of the observed VIVs of the investigated full-scale bridge with large amplitude occurred under the wind with relative low turbulence intensity (less than 8%) and wind direction nearly perpendicular to the bridge, and the amplitude of VIVs with the wind condition of high turbulence intensity was small.

Knowing that the wind field along the span-wise direction of the bridge is inhomogeneous, it is necessary to investigate the effects of the inhomogeneous wind characteristics on VIV amplitudes. Fig. 18 shows the relationship between the differences in wind characteristics at the 1/4 and 3/4 spans and RMS of vertical displacement for the VIV events under the low turbulent intensity wind (less than 8%). It can be observed that most of VIVs with large amplitude occurred in the wind condition that both difference of the inflow velocity and difference of turbulence intensity between 1/4 and 3/4 spans were small, while amplitudes of VIVs with the large difference in turbulence intensity between 1/4 and 3/4 spans were small. Above phenomenon implied that the inhomogeneity of the wind field along the span-wise direction

had a significant influence on the VIV of the full-scale bridge. It is known that vibration amplitude of VIV is related with a high correlation of wind pressures along the span-wise direction. However, the inhomogeneous characteristics along the span-wise direction of the investigated bridge lessen the correlation of wind pressures along the bridge. Thus the winds with large difference in turbulence intensity between 1/4 and 3/4 spans would induce small amplitude of VIV.

5.2. Characteristics of vortex-induced vibrations

When the vortex-shedding frequency is close to the natural frequency of the bridge deck, VIVs occur. The vortex shedding frequency is then controlled by the oscillations, and the VIV process enters into lock-in range. The relationship between the RMS of the amplitude of vertical vibrations and the reduced wind velocity obtained from the 37 VIVs is shown in Fig. 19. The damping ratio of the 37 VIVs is larger than 0.39%, as listed in Table 5. It can be seen that the maximum RMS vibration amplitude

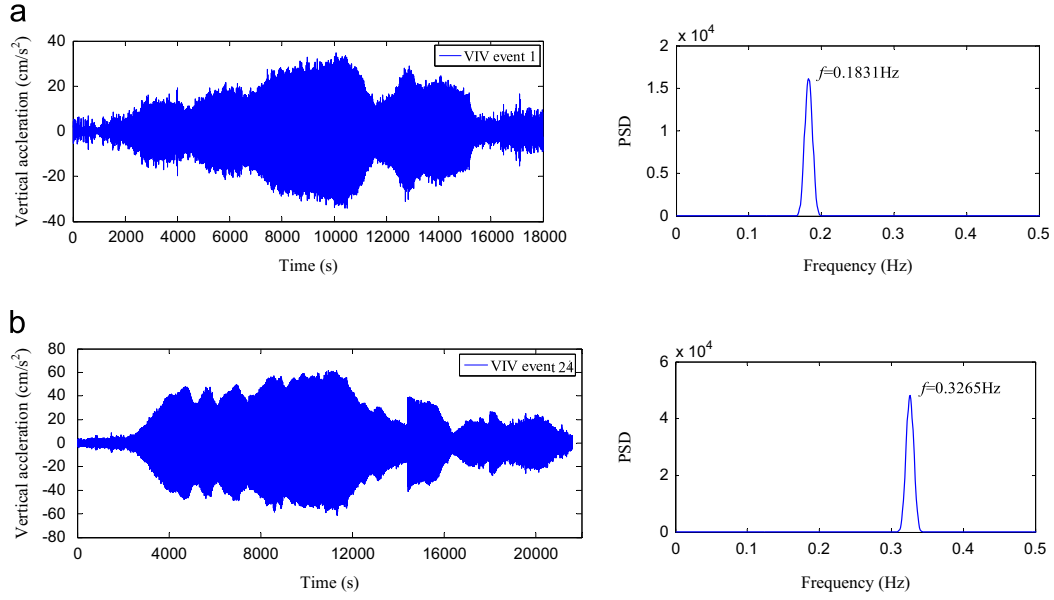


Fig. 16. Time histories of vertical accelerations and corresponding PSDs: (a) VIV event 1 with a frequency of 0.1831 Hz and (b) VIV event 24 with a frequency of 0.3265 Hz.

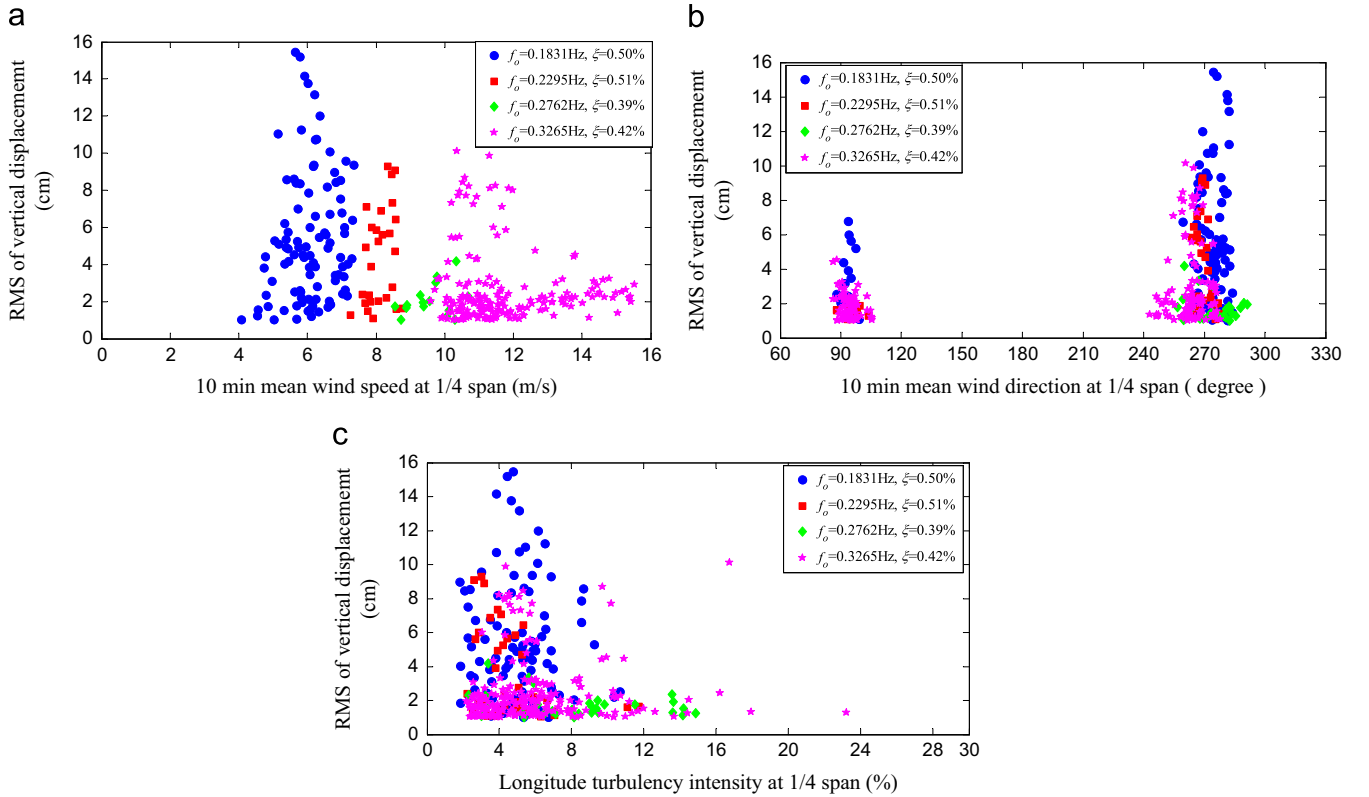


Fig. 17. Relationships between wind characteristics at the 1/4 span and the RMS of vertical displacement of the bridge deck for all the VIV events that the displacement contained one single frequency only: (a) 10-min mean wind speed, (b) 10-min mean wind direction, and (c) turbulence intensity.

occurs at $U_r=8.83$ with a value of 0.044 ($\text{RMS}(y)/D$). The lock-in range is [7.33 12.53].

For comparison, the result of wind tunnel testing on the 1:25 scale section model with a damping ratio of 0.30% is also shown in Fig. 19. The dynamic system of 1:25 section model was designed to simulate vertical mode 4 ($f=0.1831$ Hz) and the first torsional mode ($f=0.2291$ Hz) of the full-scale girder. The vertical and torsional VIVs were observed in the ranges of $U_r=[8.00\ 9.79]$ and $[11.25\ 14.89]$, respectively. The maximum RMS amplitudes of the vertical and torsional vibrations were $0.0081(\text{RMS}(y)/D)$ at

$U_r=8.73$ and 0.14° at $U_r=13.15$, respectively. Comparing the VIVs of 1:25 scale section model with the full-scale bridge, it can be observed that there have large discrepancies between the wind tunnel results and field monitoring results. First, for the full-scale bridge deck, torsional VIVs did not occur; however, they appeared in the wind tunnel test. Second, the amplitudes of the vertical VIVs of wind tunnel tests were much smaller than those in the field monitoring results.

The deviations between the wind tunnel test results and the field monitoring results are attributed to two primary reasons.

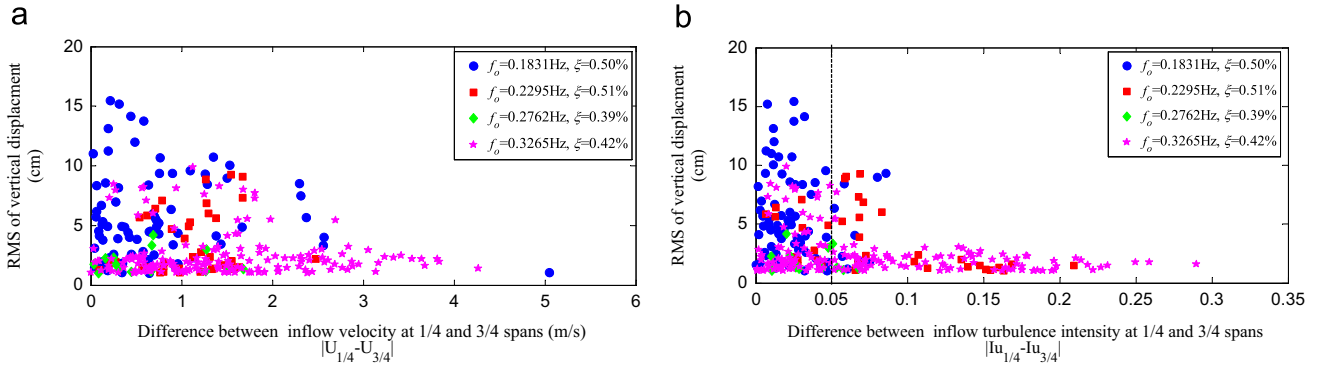


Fig. 18. Influence of inhomogeneity of the wind field along the span-wise direction on the RMS of VIV vertical displacement for the VIV events under the low turbulent intensity wind (less than 8%): (a) difference in the 10-min mean wind speed and (b) difference in turbulence intensity.

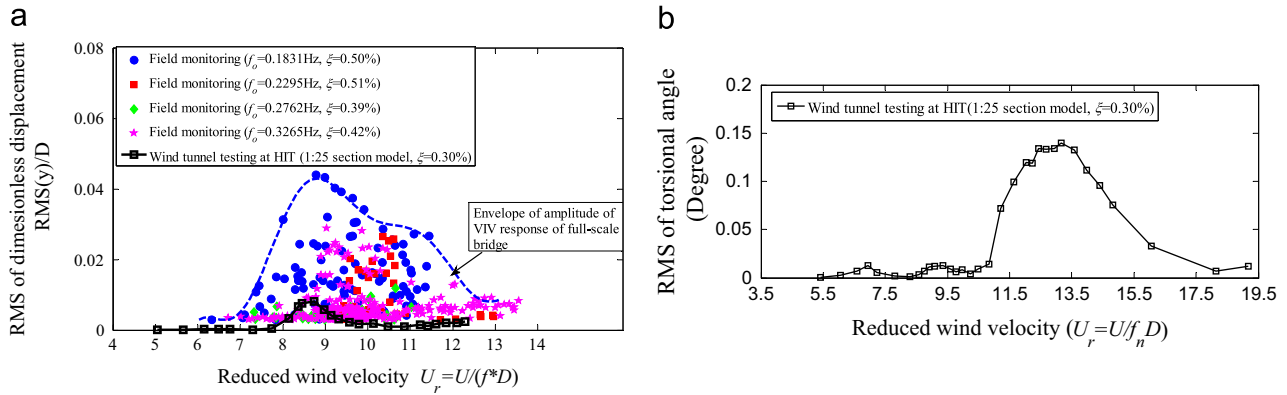


Fig. 19. VIV responses vs. the reduced wind velocity: (a) vertical VIV and (b) torsional VIV.

One reason may be attributed to that wind tunnel tests did not completely simulate the full-scale bridge and wind conditions. For example, the geometric sizes of attachments, which would induce flow separations, were too small to accurately simulate; and the flow in the wind tunnel did not satisfy two-dimension flow condition. The other may be attributed to the Reynolds number effect. For the VIVs of the section model in the wind tunnel, the Reynolds number based on the width of bridge deck is in the range of $[3.50 \times 10^5 \text{ } 5.95 \times 10^5]$. However, the Reynolds number is in the range of $[1.20 \times 10^7 \text{ } 2.88 \times 10^7]$ for the full-scale bridge. The effects of Reynolds number on the VIVs of the bridge will be further studied in the near future.

6. Conclusions

A wind and wind effect monitoring system was set up to investigate the wind-induced vibrations of a full-scale long-span suspension bridge with a twin-box girder. The wind characteristics and VIVs of this bridge were analyzed based on the monitoring data. The following conclusions were obtained.

The wind field characteristics during the summer season were analyzed. It was observed that the wind around the 3/4 span had a lower mean wind speed and much stronger turbulence intensity than that around the 1/4 span, which indicated that the wind field along the span-wise direction was inhomogeneous. The inhomogeneity was resulted in by the geography of the surroundings nearby the bridge.

The full-scale pressure distribution of the lower surface was monitored and analyzed. A field flow visualization test was performed. The mean pressure coefficient reached its maximum suction value around the front rail and then became extremely

small rapidly. The fluctuating pressure coefficient showed high fluctuations around the corner. The Strouhal number of the full-scale twin-box girder is 0.1159, which was obtained according to the PSDs of pressures on the trailing edge of the lower surface. Combining the PSDs of the wind pressures and flow visualization, it can be concluded that the flow separation on the lower surface of the tail, which was around the windward corner, caused large coherent vortex shedding at the leading edge. The vortex traveled downstream and merged with vortices generated at the trailing edge of the downstream box girder, and finally, alternating shedding Karman-like vortices formed at the tail of downstream box girder.

A comparison between the 1:25 section model and field monitoring results was made in this paper. The results show that section model had higher suction than the full-scale bridge in the separation region, and the Strouhal number obtained from the section model was smaller, which implied that the section model test overestimated flow separation.

Thirty-seven vertical VIV events with frequencies of 0.1831, 0.2295, 0.2734 and 0.3265 Hz were observed by the field monitoring system. For these VIV events, the maximum amplitude of displacement was 23.89 cm with a frequency of 0.1831 Hz, and the maximum amplitude of acceleration was 61.82 cm/s^2 with a frequency of 0.3265 Hz. For the real bridge, it is observed that VIVs of field full-scale bridge occurred in the wind conditions with low turbulence intensity and wind direction nearly perpendicular to the bridge. Most of the VIVs with large amplitude occurred in the wind condition that differences of the inflow velocity and turbulence intensity between 1/4 and 3/4 spans were small, which implied that the inhomogeneity of the wind field along the span-wise direction had a significant effect on the VIV of the full-scale bridge. Comparison of the section model testing result and field

monitoring result indicated that the amplitude of the vertical VIV of the section model was much smaller than that of the full-scale bridge, which may be attributed to the Reynolds number effects and flow pattern and need to be further investigated in the future.

Acknowledgments

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